

[ASRJC/2025/Q3]

3 Name your Jupyter Notebook as:

TASK3_<your name>_<centre number>_<index number>.ipynb

A H2 Computing teacher has a collection of fill-in-the-blanks problems that needs to be managed using a MongoDB database.

For each of the sub-tasks, add a comment statement at the beginning of the code using the symbol '#', to indicate the sub-task the program code belongs to, for example:

In [1]:

```
#Task 3.1
Program code
```

Output:

Task 3.1

The file `problems.txt` contains problems separated by an empty line.

Write program code to extract the data from the file and store them in a MongoDB collection named `problems` within a database named `computing`.

Note that the answers to the blanks in a problem should be stored as a list of strings, and the attempt count stored as an integer value. [5]

Task 3.2

Write MongoDB query logic to extract problems that meet both of the following criteria:

- the topic is either 'Data Structures' or 'Algorithms'
- the number of attempts for the problems is between 5 and 10 (both inclusive).

Display the question texts of these matching problems. [3]

Task 3.3

Since the topic 'Socket Programming' will no longer be there in the revised H2 Computing syllabus, the teacher wants to update the collection by removing all problems tagged with that topic.

Write MongoDB logic to perform this update and execute it. [2]

Task 3.4

Write program code that:

- randomly selects a problem from the MongoDB collection
- displays the topic and the question text of the selected problem
- prompts the student to enter an answer for each blank in the question, reminding him to include the first letter in his response
- checks the student's answers against the expected answers (case-insensitive)
- outputs summative results (e.g. "You got 3 out of 5 answers correct!"), without revealing what the correct answers are
- increments the attempt count for that problem by 1 in the database.

If the student's answers to the problem are not all correct, prompt the student to enter all the answers again for the same problem. This continues until the student answers the entire problem correctly.

Test your code appropriately. [7]

Save your Jupyter Notebook for Task 3.